

Positive Strongly Kreiss Weighted Shifts Have Polylogarithmic Power Growth

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Status: partial result, likely valid; human review requested.

1 Source question

Arnold and Cuny ask the following in Question 5.8 of [1]: if $1 < p < \infty$ and T is a positive strongly Kreiss bounded operator on an L^p -space, must there be $C, \kappa > 0$ such that

$$\|T^N\| \leq C \log^\kappa(N+1) \quad (N \in \mathbb{N})?$$

The source question is reproduced below from page 19 of the arXiv PDF.

If X is an (AM)-space, then X^* is an (AL)-space. It is known that an (AL)-space is isometrically isomorphic to some L^1 space and that an (AM)-space is isometrically isomorphic to some $C(K)$ where K is a compact space. We are now ready to prove the following statement.

Proposition 5.6. *Let T be a positive strongly Kreiss bounded operator on an (AL)-space or an (AM)-space. Then there exist $C, \kappa > 0$ such that for every $N \in \mathbb{N}$,*

$$(5.8) \quad \|T^N\| \leq C \log^\kappa(N+1).$$

Remark 5.7. Since $L^\infty(\mathcal{M})$ is an (AM)-space, Proposition 5.2 remains valid also for the case $p = \infty$ by the above result.

Proof. If T is a positive strongly Kreiss bounded operator on an (AL)-space, then using (5.7), it is straightforward that T is absolutely strongly Kreiss bounded, and we can then conclude invoking Lemma 5.1.

If T is a positive strongly Kreiss bounded operator on an (AM)-space, then T^* is a positive strongly Kreiss bounded on an (AL)-space. Then this case is reduced to the first part of the proof. $\square$

Question 5.8. In view of the considerations in Section 5, it is natural to ask whether the bound in Proposition 5.2 can be improved to obtain a logarithmic bound. More precisely, for a positive strongly Kreiss bounded operator T on $L^p(\mathcal{M})$ with $1 < p < \infty$, are there $C, \kappa > 0$ such that for every $N \in \mathbb{N}$,

$$\|T^N\| \leq C \log^\kappa(N+1) ?$$

We prove an affirmative answer for positive unilateral weighted backward shifts whose weights are at least one. This class includes the familiar telescoping logarithmic examples discussed in [1, 2].

2 Definitions

Let $1 \leq p \leq \infty$ and use the convention $\mathbb{N}_0 = \{0, 1, 2, \dots\}$. Given a bounded sequence $(w_j)_{j \geq 0}$ of positive numbers, the unilateral weighted backward shift T on $\ell^p(\mathbb{N}_0)$ is

$$(Tx)_j = w_j x_{j+1}.$$

Put $a_0 = 1$ and $a_n = \prod_{j=0}^{n-1} w_j$ for $n \geq 1$. Then

$$(T^m x)_j = \frac{a_{j+m}}{a_j} x_{j+m}, \quad \|T^m\| = \sup_{j \geq 0} \frac{a_{j+m}}{a_j}. \quad (1)$$

The strong Kreiss condition is the iterated resolvent estimate from [1]. We use its equivalent exponential formulation: there is $K_{\text{SK}} \geq 1$ such that

$$\|e^{zT}\| \leq K_{\text{SK}} e^{|z|} \quad (z \in \mathbb{C}). \quad (2)$$

This equivalence is recalled in the introduction of [1].

3 Proof intuition

For a positive shift, $e^{-N} e^{NT}$ is a Poisson average of its powers. Poisson mass is of order $N^{-1/2}$ throughout a window of width of order $\sqrt{N}$ around N . Test this average on a constant vector supported on one coordinate block of that width and inspect its image on a second block. Monotonicity of the cumulative weights makes every matrix entry between the two blocks at least a fixed Poisson factor times $a_{n+N}/a_{n+c\sqrt{N}}$. There are $c\sqrt{N}$ entries in every relevant row, exactly cancelling the $N^{-1/2}$ Poisson scale in every ℓ^p norm.

The strong Kreiss bound therefore forces the cumulative weight to grow by only a fixed factor when its index is reduced from N to $c\sqrt{N}$. Iterating square roots takes only $O(\log \log N)$ steps, so the total growth is at most a power of $\log N$. The remaining general positive-operator question is open because the coordinate-block test uses the shift geometry.

4 The partial theorem

Theorem 1. *Let $1 \leq p \leq \infty$, and let T be the weighted backward shift on $\ell^p(\mathbb{N}_0)$ defined by a bounded sequence (w_j) satisfying $w_j \geq 1$. If T is strongly Kreiss bounded, then there are constants $C, \kappa > 0$ such that*

$$\|T^N\| \leq C \log^\kappa(N+2) \quad (N \geq 1).$$

In particular, Question 5.8 has an affirmative answer in this class.

We first record the only scalar estimate needed.

Lemma 2 (Poisson window). *There are constants $\eta, c_0 > 0$ and $N_0 \in \mathbb{N}$ such that, whenever $N \geq N_0$ and $m \in \mathbb{N}_0$ satisfies $|m - N| \leq \eta\sqrt{N}$, one has*

$$e^{-N} \frac{N^m}{m!} \geq \frac{c_0}{\sqrt{N}}.$$

Proof. By the upper Stirling estimate, for an absolute $C > 0$,

$$m! \leq C\sqrt{m} (m/e)^m.$$

Writing $m = N + s$, the exponential factor in the resulting lower bound is

$$-N + m + m \log(N/m) = s - (N + s) \log(1 + s/N).$$

For $|s| \leq \eta\sqrt{N}$ this expression is bounded below by an absolute constant, while m is comparable with N . Decreasing c_0 handles the finitely many remaining values after fixing N_0 . $\square$

Lemma 3 (Square-root recurrence). *Under the assumptions of Theorem 1, define*

$$b_N := \|T^N\| = \sup_{n \geq 0} \frac{a_{n+N}}{a_n}.$$

There are $C_0 \geq 1$, $c \in (0, 1)$, and $N_1 \in \mathbb{N}$ such that

$$b_N \leq C_0 b_{\lfloor c\sqrt{N} \rfloor} \quad (N \geq N_1). \quad (3)$$

Proof. Let η, c_0, N_0 be as in Lemma 2. For large N , put $L = \lfloor \eta\sqrt{N}/2 \rfloor$. From (2), the normalized Poisson operator

$$P_N := e^{-N} e^{NT} = \sum_{m=0}^{\infty} e^{-N} \frac{N^m}{m!} T^m$$

satisfies $\|P_N\| \leq K_{\text{SK}}$.

Fix $n \geq 0$. If $p < \infty$, let

$$x = L^{-1/p} \sum_{k=n+N}^{n+N+L-1} e_k;$$

if $p = \infty$, omit the factor $L^{-1/p}$. In either case $\|x\|_p = 1$. For $j \in \{n, \dots, n+L-1\}$ and every $k \in \text{supp } x$, the integer $m = k - j$ satisfies $|m - N| < L$, and hence its Poisson coefficient is at least $c_0/\sqrt{N}$. Since $w_j \geq 1$, the sequence (a_j) is nondecreasing, and therefore

$$\frac{a_k}{a_j} \geq \frac{a_{n+N}}{a_{n+L}}.$$

Using (1), all terms are nonnegative and we obtain

$$(P_N x)_j \geq \frac{c_0 L}{\sqrt{N}} \frac{a_{n+N}}{a_{n+L}} L^{-1/p}.$$

Taking the ℓ^p norm over the L displayed output coordinates (with the usual interpretation for $p = \infty$), and using $L/\sqrt{N} \geq \eta/4$ for large N , gives

$$K_{\text{SK}} \geq \|P_N x\|_p \geq \frac{c_0 \eta}{4} \frac{a_{n+N}}{a_{n+L}}.$$

Thus $a_{n+N}/a_{n+L} \leq C_0 := 4K_{\text{SK}}/(c_0\eta)$, uniformly in n . Consequently,

$$\frac{a_{n+N}}{a_n} = \frac{a_{n+N}}{a_{n+L}} \frac{a_{n+L}}{a_n} \leq C_0 b_L.$$

Taking the supremum over n proves (3), after changing c and the finite threshold if necessary. $\square$

Proof of Theorem 1. The numbers b_N are nondecreasing because (a_j) is nondecreasing. Starting from N , repeatedly apply Lemma 3 while the current index is at least N_1 . The indices obey $N_{r+1} \leq c\sqrt{N_r}$, so the process stops after at most

$$r \leq C_1 + \log_2 \log(N + 2)$$

steps, where C_1 depends only on c and N_1 . If $B = \max_{1 \leq m \leq N_1} b_m$, iteration yields

$$b_N \leq B C_0^r \leq C \log(N + 2)^{\log_2 C_0}.$$

This proves the theorem with $\kappa = \log_2 C_0$ (or any larger positive number). $\square$

5 Verification and limitations

The proof is independent of computation. The included script `code/weighted_shift_probe.py` computes finite-section induced norms of $e^{-r}e^{rT}$ for cumulative weights $a_n = (n+1)^\beta$, three values of p , three values of β , and four radii. The observed norms increase with the radius, as predicted by the theorem because polynomial power growth cannot coexist with a uniform strong Kreiss bound in this monotone shift class. This finite computation is only a sanity check.

The result does not cover arbitrary positive operators on L^p , nor weighted shifts with non-monotone cumulative products. A bounded literature search through 19 July 2026 used the phrases “positive strongly Kreiss bounded”, “strongly Kreiss weighted shift”, and “weighted shift logarithmic powers”, and checked [1, 2, 3]. No statement of Lemma 3 or Theorem 1 was found. Novelty confidence is moderate because the argument may be folklore in weighted-shift theory.

The recommended human-review focus is the uniform Poisson lower bound and the two block endpoints in Lemma 3. Once that lemma is accepted, the logarithmic iteration is immediate.

References

- [1] L. Arnold and C. Cuny, *On the growth rate of powers of a strongly Kreiss bounded operator on an L^p -space*, Studia Math., published online 23 March 2026, DOI: [10.4064/sm230303-29-10](https://doi.org/10.4064/sm230303-29-10); arXiv:2302.14135.
- [2] G. Cohen, C. Cuny, T. Eisner, and M. Lin, *Resolvent conditions and growth of powers of operators*, J. Math. Anal. Appl. 487 (2020), 124035; arXiv:1912.10507.
- [3] T. Bermudez, A. Bonilla, V. Muller, and A. Peris, *Cesaro bounded operators in Banach spaces*, J. Math. Anal. Appl. 472 (2019), 1264–1278; arXiv:1706.03638.